Newton Interpolation

- Consider our data set of n+1 points $y_i=f(x_i)$ at $x_0,x_1 \ldots x_i, \ldots x_n$: $x_n > x_0$
- Since $p_n(x)$ is the unique polynomial $p_n(x)$ of order n, write it:

$$p_n(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + \cdots + b_n(x - x_0)\cdots(x - x_{n-1})$$

$$b_0 = f(x_0)$$

$$b_1 = f[x_1, x_0] = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_2 = f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0}$$

$$\vdots$$

$$b_n = f[x_n, x_{n-1}, \ldots, x_0] = \frac{f[x_n, \ldots, x_1] - f[x_{n-1}, \ldots, x_0]}{x_n - x_0}$$

- $f[x_i,x_j]$ is a first divided difference
- $f[x_2,x_1,x_0]$ is a second divided difference, etc.

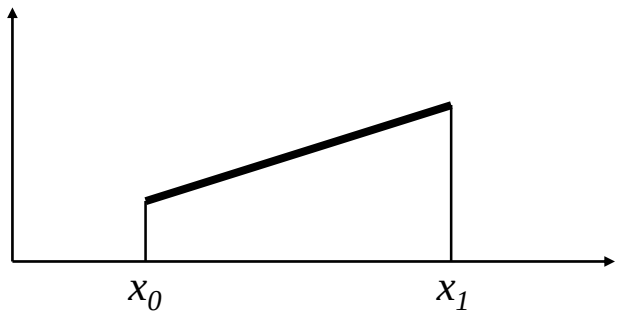
Invariance Theorem

- Note, that the order of the data points does not matter.
- All that is required is that the data points are distinct.
- Hence, the divided difference $f[x_0, x_1, \ldots, x_k]$ is invariant under all permutations of the x_i‘s.

Linear Interpolation

- Simple linear interpolation results from having only 2 data points.

$$p_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$



Quadratic Interpolation

- Three data points:

$$p_2(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

$$= f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0) + \frac{\left[\frac{f(x_2) - f(x_1)}{x_2 - x_1}\right] - \left[\frac{f(x_1) - f(x_0)}{x_1 - x_0}\right]}{x_2 - x_0}(x - x_0)(x - x_1)$$

$$= f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$
$$+ \frac{\left(\left[\frac{f(x_2) - f(x_1)}{x_2 - x_1}(x - x_1)\right](x - x_0)\right) - \left(\left[\frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)\right](x - x_1)\right)}{x_2 - x_0}$$

Newton Interpolation

- Let's look at the recursion formula:

$$b_n = f[x_n, x_{n-1}, \dots, x_0] = \frac{f[x_n, \dots, x_1] - f[x_{n-1}, \dots, x_0]}{x_n - x_0}$$

- For the quadratic term:

$$b_2 = f[x_2, x_1, x_0] = \frac{f[x_2, x_1] - f[x_1, x_0]}{x_2 - x_0} = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

$$= \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - b_1}{x_2 - x_0}$$

Evaluating for x_2

$$f(x_2) = b_0 + b_1(x_2 - x_0) + b_2(x_2 - x_0)(x_2 - x_1)$$

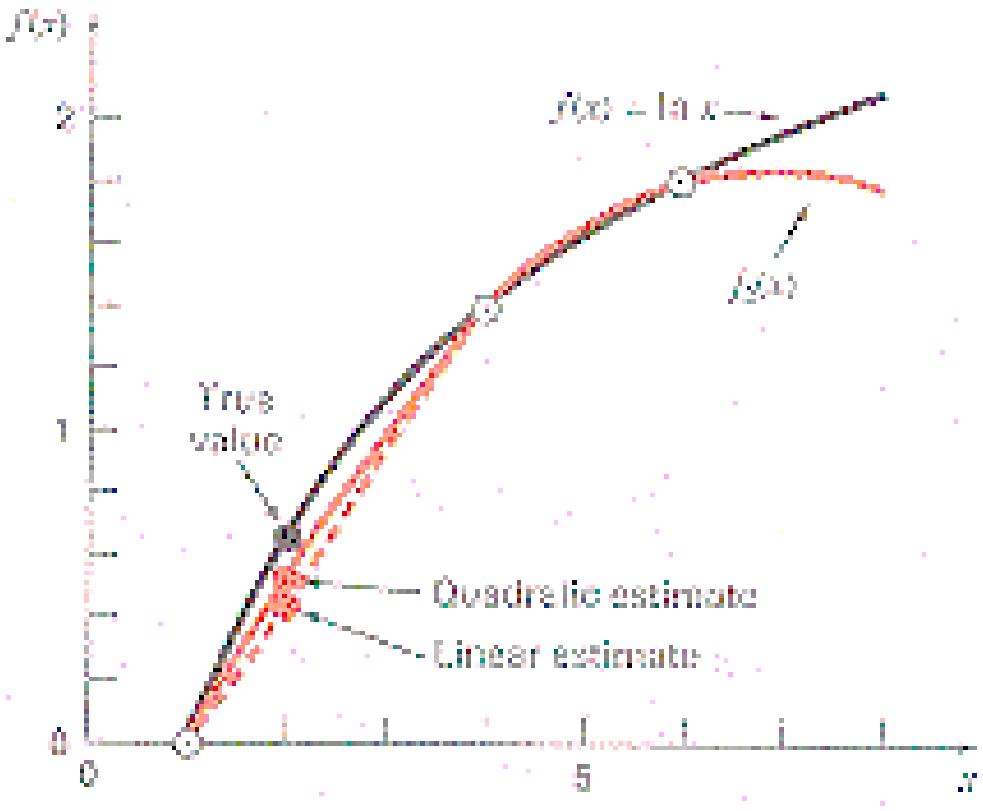
$$= f_0 + b_1(x_2 - x_0) + \left(\frac{f_2 - f_1}{x_2 - x_1} - b_1\right)(x_2 - x_1)$$

$$= f_0 + \left(\frac{f_1 - f_0}{x_1 - x_0}\right)(x_1 - x_0) + f_2 - f_1$$

$$= f_2$$

Example: ln(x)

- Interpolation of ln(2): given ln(1); ln(4) and ln(6)
  - Data points: {(1,0), (4,1.3863), (6,1.79176)}
  - Linear Interpolation:
  - 0 + {(1.3863-0)/(4-1)}(x-1) = 0.4621(x-1)
  - Quadratic Interpolation:
  - 0.4621(x-1)+((0.20273-0.4621)/5)(x-1)(x-4)
  - = 0.4621(x-1) - 0.051874 (x-1)(x-4)



Example: ln(x)

- Quadratic interpolation catches some of the curvature
- Improves the result somewhat
- Not always a good idea: see later…

Calculating the Divided-Differences

- A *divided-difference* table can easily be constructed incrementally.
- Consider the function *ln(x)*.

x	ln(x)	f[I,I+1]
1	**0.000000**	
2	0.693147	**0.693147**
3	1.098612	0.405465
4	1.386294	0.287682
5	1.609438	0.223144
6	1.791759	0.182322
7	1.945910	0.154151
8	2.079442	0.133531
x	ln(x)	b10-b9)/(A10-A9

Calculating the Divided-Differences

x	ln(x)	f[I,I+1]	
1	**0.000000**		
2	0.693147	**0.693147**	
3	1.098612	0.405465	**-0.143841**
4	1.386294	0.287682	-0.058892
5	1.609438	0.223144	-0.032269
6	1.791759	0.182322	-0.020411
7	1.945910	0.154151	-0.014085
8	2.079442	0.133531	-0.010310
x	ln(x)	b10-b9)/(A10-A9	(c10-c9)/(a10-a8)

$$f[i.i+1] = \frac{f[x_{i+1}] - f[x_i]}{(x_{i+1} - x_i)}$$

Calculating the Divided-Differences

x	ln(x)	f[I,I+1]		
1	**0.000000**			
2	0.693147	**0.693147**		
3	1.098612	0.405465	**-0.143841**	
4	1.386294	0.287682	-0.058892	**0.028317**
5	1.609438	0.223144	-0.032269	0.008874
6	1.791759	0.182322	-0.020411	0.003953
7	1.945910	0.154151	-0.014085	0.002109
8	2.079442	0.133531	-0.010310	0.001259
x	ln(x)	b10-b9)/(A10-A9	(c10-c9)/(a10-a8)	d10-d9)/(a10-a7

$$f[i.i+1,i+2] = \frac{f[i+1,i+2] - f[i,i+1]}{(x_{i+2} - x_i)}$$

Calculating the Divided-Differences

x	ln(x)	f[I,I+1]			
1	**0.000000**				
2	0.693147	**0.693147**			
3	1.098612	0.405465	**-0.143841**		
4	1.386294	0.287682	-0.058892	**0.028317**	
5	1.609438	0.223144	-0.032269	0.008874	**-0.004861**
6	1.791759	0.182322	-0.020411	0.003953	-0.001230
7	1.945910	0.154151	-0.014085	0.002109	-0.000461
8	2.079442	0.133531	-0.010310	0.001259	-0.000212
x	ln(x)	b10-b9)/(A10-A9	(c10-c9)/(a10-a8	d10-d9)/(a10-a7	d10-d9)/(a10-a6

$$f[i, \dots, i+3] = \frac{f[i+1, i+2, i+3] - f[i, i+1, i+2]}{(x_{i+3} - x_i)}$$

Calculating the Divided-Differences

x	ln(x)	f[I,I+1]					
1	**0.000000**						
2	0.693147	**0.693147**					
3	1.098612	0.405465	**-0.143841**				
4	1.386294	0.287682	-0.058892	**0.028317**			
5	1.609438	0.223144	-0.032269	0.008874	**-0.004861**		
6	1.791759	0.182322	-0.020411	0.003953	-0.001230	**0.000726**	
7	1.945910	0.154151	-0.014085	0.002109	-0.000461	0.000154	
8	2.079442	0.133531	-0.010310	0.001259	-0.000212	0.000050	
x	ln(x)	b10-b9)/(A10-A9	c10-c9)/(a10-a8	d10-d9)/(a10-a7	d10-d9)/(a10-a6	e10-e9)/(a10-a5	[redacted]

$$f[i, \ldots, i+4] = \frac{f[i+1, \ldots, i+4] - f[i, \ldots, i+3]}{(x_{i+4} - x_i)}$$

Calculating the Divided-Differences

x	ln(x)	f[I,I+1]						
1	**0.000000**							
2	0.693147	**0.693147**						
3	1.098612	0.405465	**-0.143841**					
4	1.386294	0.287682	-0.058892	**0.028317**				
5	1.609438	0.223144	-0.032269	0.008874	**-0.004861**			
6	1.791759	0.182322	-0.020411	0.003953	-0.001230	**0.000726**		
7	1.945910	0.154151	-0.014085	0.002109	-0.000461	0.000154	**-0.000095**	
8	2.079442	0.133531	-0.010310	0.001259	-0.000212	0.000050	-0.000017	
x	ln(x)	b10-b9)/(A10-A9	(c10-c9)/(a10-a8	d10-d9)/(a10-a7	d10-d9)/(a10-a6	e10-e9)/(a10-a5	f10-f9)/(a10-a4	

$$f[i, \dots, i+5] = \frac{f[i+1, \dots, i+5] - f[i, \dots, i+4]}{(x_{i+5} - x_i)}$$

Calculating the Divided-Differences

x	ln(x)	f[I,I+1]						f[I,I+1,…,I+7]
1	**0.000000**							
2	0.693147	**0.693147**						
3	1.098612	0.405465	**-0.143841**					
4	1.386294	0.287682	-0.058892	**0.028317**				
5	1.609438	0.223144	-0.032269	0.008874	**-0.004861**			
6	1.791759	0.182322	-0.020411	0.003953	-0.001230	**0.000726**		
7	1.945910	0.154151	-0.014085	0.002109	-0.000461	0.000154	**-0.000095**	
8	2.079442	0.133531	-0.010310	0.001259	-0.000212	0.000050	-0.000017	**0.000013**
x	ln(x)	b10-b9)/(A10-A9	(c10-c9)/(a10-a8	d10-d9)/(a10-a7	d10-d9)/(a10-a6	e10-e9)/(a10-a5	f10-f9)/(a10-a4	(g10-g9)/(a10-a3)

$$f[i,\ldots,i+6] = \frac{f[i+1,\ldots,i+6] - f[i,\ldots,i+5]}{(x_{i+6} - x_i)}$$

Calculating the Divided-Differences

- Finally, we can calculate the last coefficient.

x	ln(x)	f[I,I+1]						f[I,I+1,…,I+7]
1	**0.000000**							
2	0.693147	**0.693147**						
3	1.098612	0.405465	**-0.143841**					
4	1.386294	0.287682	-0.058892	**0.028317**				
5	1.609438	0.223144	-0.032269	0.008874	**-0.004861**			
6	1.791759	0.182322	-0.020411	0.003953	-0.001230	**0.000726**		
7	1.945910	0.154151	-0.014085	0.002109	-0.000461	0.000154	**-0.000095**	
8	2.079442	0.133531	-0.010310	0.001259	-0.000212	0.000050	-0.000017	**0.000011**
x	ln(x)	b10-b9)/(A10-A9	(c10-c9)/(a10-a8	d10-d9)/(a10-a7	d10-d9)/(a10-a6	e10-e9)/(a10-a5	f10-f9)/(a10-a4	(g10-g9)/(a10-a3)

$$f[i,\ldots,i+7] = \frac{f[i+1,\ldots,i+7] - f[i,\ldots,i+6]}{(x_{i+7} - x_i)}$$

Calculating the Divided-Differences

- All of the coefficients for the resulting polynomial are in bold.

x	ln(x)	f[I,I+1]						f[I,I+1,…,I+7]
1	**0.000000**							
2	0.693147	**0.693147**						
3	1.098612	0.405465	**-0.143841**					
4	1.386294	0.287682	-0.058892	**0.028317**				
5	1.609438	0.223144	-0.032269	0.008874	**-0.004861**			
6	1.791759	0.182322	-0.020411	0.003953	-0.001230	**0.000726**		
7	1.945910	0.154151	-0.014085	0.002109	-0.000461	0.000154	**-0.000095**	
8	2.079442	0.133531	-0.010310	0.001259	-0.000212	0.000050	-0.000017	**0.000011**
x	ln(x)	b10-b9)/(A10-A9	(c10-c9)/(a10-a8	d10-d9)/(a10-a7	d10-d9)/(a10-a6	e10-e9)/(a10-a5	f10-f9)/(a10-a4	(g10-g9)/(a10-a3)

Polynomial Form for Divided-Differences

- The resulting polynomial comes from the divided-differences and the corresponding product terms:

$$p_7(x) = 0 + 0.693(x-1) - 0.144(x-1)(x-2) + 0.28(x-1)(x-2)(x-3)$$

$$-0.0049(x-1)(x-2)(x-3)(x-4)$$

$$+7.26 \times 10^{-4}(x-1)(x-2)(x-3)(x-4)(x-5)$$

$$-9.5 \times 10^{-5}(x-1)(x-2)(x-3)(x-4)(x-5)(x-6)$$

$$+1.1 \times 10^{-5}(x-1)(x-2)(x-3)(x-4)(x-5)(x-6)(x-7)$$

Many polynomials

- Note, that the order of the numbers (x_i, y_i)'s only matters when writing the polynomial down.
  - The first column represents the set of linear splines between two adjacent data points.
  - The second column gives us quadratics thru three adjacent points.
  - Etc.

Adding an Additional Data Point

- Adding an additional data point, simply adds an additional term to the existing polynomial.
  - Hence, only n additional divided-differences need to be calculated for the $n+1^{\underline{st}}$ data point.

x	ln(x)	f[I,I+1]					f[I,I+1,…,I+7]		
1.0000000	**0.0000000**								
2.0000000	0.6931472	**0.6931472**							
3.0000000	1.0986123	0.4054651	**-0.1438410**						
4.0000000	1.3862944	0.2876821	-0.0588915	**0.0283165**					
5.0000000	1.6094379	0.2231436	-0.0322693	0.0088741	**-0.0048606**				
6.0000000	1.7917595	0.1823216	-0.0204110	0.0039528	-0.0012303	**0.0007261**			
7.0000000	1.9459101	0.1541507	-0.0140854	0.0021085	-0.0004611	0.0001539	**-0.0000954**		
8.0000000	2.0794415	0.1335314	-0.0103096	0.0012586	-0.0002125	0.0000497	-0.0000174	**0.0000111**	
1.5000000	0.4054651	0.2575348	-0.0225461	0.0027192	-0.0004173	0.0000819	-0.0000215	0.0000082	**-0.0000058**

Adding More Data Points

- Quadratic interpolation:
  - does linear interpolation
  - Then add higher-order correction to catch the curvature
- Cubic, …
- Consider the case where the data points are organized such the first two are the endpoints, the next point is the mid-point, followed by successive mid-points of the half-intervals.
  - Worksheet: $f(x)=x^2$ from -1 to 3.

Error

- Define the error term as:

$$\varepsilon_n(x) = f(x) - p_n(x)$$

- If $f(x)$ is an $n^{\underline{th}}$ order polynomial $p_n(x)$ is of course exact.
- Otherwise, since there is a perfect match at $x_0, x_1, \ldots, x_n$
- This function has at least n+1 roots at the interpolation points.

$$\therefore \varepsilon_n(x) = (x - x_0)(x - x_1) \cdots (x - x_n)h(x)$$

Example

- Suppose f was a polynomial of degree m, where $m<n$.
- Ex: $f(x) = 3x^2 - x + 2$
- We have evaluations of $f(x)$ at five locations:
- (-2, 12), (-1, 4), (0, 2), (1, 4), (2, 13)
- Determine *$f(0.5)$ and error at $x=0.5$*